

LANDASSIST

AI-POWERED EARTH OBSERVATION & VECTOR DELIVERY SYSTEM FOR OPENSTREETMAP

TARGET AOI: KABUPATEN TANGERANG (1,193 KM²) · FOUNDATION MODEL: GOOGLE SATELLITE EMBEDDINGS 64-BAND · DEEP LEARNING: SEGFORMER MIT-B2 · CLOUD VECTOR ENGINE: POSTGIS + FASTAPI

MAKING MAPS TOGETHER WITH MULTIMODAL GEOSPATIAL MACHINE LEARNING · OPENSTREETMAP-COMPLIANT FEATURE DELIVERY

1,193 km²
REGIONAL COVERAGE
Entire county boundary mapped via 2×2 grid tile partitioning

64-Band
MULTIMODAL GSE
Google Satellite Embeddings @ 10m GSD (Sentinel-2 annual composite)

70.86%
SEGFORMER MIOU
Test evaluation on residential & industrial ground truth labels

< 200 ms
VECTOR DELIVERY
Sub-second spatial queries via PostGIS GIST index & streaming GeoJSON

01 CHALLENGE & AUTONOMOUS GEOSPATIAL VISION

Traditional mapping of rapidly urbanizing regions like **Kabupaten Tangerang (~1,193 km²)** requires hundreds of hours of manual remote tracing. Human digitizers face high cognitive fatigue, subjective boundary placement, and prolonged turnaround times.

LandAssist solves this with a foundation-model pipeline that converts 64-band multispectral embeddings into topologically regularized, vector-clean OpenStreetMap polygons served directly to mappers in real time.

02 END-TO-END DEEP LEARNING & VECTOR PIPELINE

STAGE 1 • ACQUISITION
Earth Engine 64-Band GSE
Partitions AOI into 4 sub-tiles (mitigating GEE 48 MB limit), acquiring annual Sentinel-2 latent embeddings at 10m ground resolution.

STAGE 2 • SEGMENTATION
SegFormer MiT-B2 Inference
Custom 64-input patch stem with 2D Hann window blending over disk memory-mapped buffers. Fast GPU T4 batched execution (~2 min).

STAGE 3 • REGULARIZATION
GDAL Sieve & Chaikin Curvature
Eliminates noise (< 1,600 m² MMU) and applies Chaikin corner-cutting in UTM Zone 48S (EPSG:32748) to smooth pixel-step staircase artifacts.

STAGE 4 • VECTOR INGESTION
PostGIS & FastAPI Vector Tiles
Streamed ingestion into Supabase PostGIS with GIST indexing; served over REST GeoJSON and Mapbox Vector Tiles (MVT) with CORS.

03 OSM LANDUSE TAXONOMY & CLASS MAPPING

LandAssist models strictly output OpenStreetMap-compliant attribute tags with lineage tracking (`source=LandAssist_v1.0.0;GSE`):

Background
`class=0`

Industrial
`landuse=industrial`

Residential
`landuse=residential`

04 STANDALONE WEB EDITOR (RENDER CLOUD 24/7)

LandAssist hosts a standalone, customized Rapid/iD Web Editor deployed in the cloud. Mappers can visualize predicted polygons seamlessly overlaid on real-world high-resolution satellite basemaps.

Key Editor Capabilities:
- Automatic Bounding Box querying via spatial REST API
- Single-click conversion into editable OSM polygon features
- Preset OSM tagging with verified confidence scores
- Zero local environment or GIS software installation needed

WORKFLOW METRIC	MANUAL DIGITIZING	LANDASSIST AI
Time (1,193 km²)	~3 to 4 Weeks	**~12 Minutes (GPU T4)**
Labor & Fatigue	High Mapper Strain	**Automated Cloud Batch**
Boundary Geometry	Subjective Variances	**Chaikin Regularized**
Vector Access	Manual Export/Upload	**Live PostGIS API**

EXPLORE LIVE AT THE EXPO

Scan directly with your smartphone to test the live web platform and inspect the open-source engineering:

LIVE WEB EDITOR
Test interactive AI polygon editing

GITHUB REPOSITORY
Inspect SegFormer weights,

Project Lead & ML Architecture: Bayu Nabil (Head of Project Management)

Dataset Annotation & Labeling: Yabes B. • Ahmad F. R. • FireLies • Naumam